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Interview with the radiologist - 27/11/2017

Multiple systems used. Carestream (built on Nuance) was shown for a X-ray exploration and AquariusNet viewer was shown for a CT scan exploration.
Data storage by PACS (with backup: data saved at the hospital and at the university) and following HL7 standards.
The input and output of the diagnosis are DICOM data files.
The usual imaging data are CT scans, X-ray images, MRI images, ultrasounds, and text data (e.g., physician report, clinical information and other hospitals' reports).

The workflow starts with a specific surgeon request regarding a given image. Then, a comparison between the current and the previous scans is performed (when possible). While analyzing the image, the radiologist started compiling the (non-structured) report by speech recognition system. The interactions involved in this process were drawing a circle saving the new image as a key image, text annotations, drawing an arrow, performing angle, density and distance measurements, and using the spine labeling tool. The past scans of the patients were retrieved and checked quickly over the analysis. Once done, the radiologist sends the new images and the report as DICOM data to the physician, and it is stored. In case of issues in understanding the analysis results, the physician can send the DICOM data with a new request for clarifying some points. However, this occurs rarely. Otherwise, the procedure is ended.

The demographic information of the patient and the date of the previous images are relevant for the analysis.
Many clicks on the mouse and the speech recorder are required for the analysis and the report.
It emerged that there is a need to visualize and link the images related to what mentioned in the report. This would help and accelerate the understanding of the report, to improve the follow ups and to reduce errors due to misunderstanding.
The reports are used as checklists. In some cases, merging report and images would be too overwhelming and confusing.
Regarding collaborative work: informal help between radiologists occurs, but it is not mentioned in the report sometimes. It is too time-consuming. A solution can be to split the analysis in two parts where a radiologist (or machine) would perform the measurements and pre-process the data, then a second one would elaborate the data in order to extract the results.
Since the main audience of the radiologist are often the physicians rather than the patients and since the communication between them is not always clear, it would be interesting to interview some clinicians who generally collaborate with radiologists.

Interview with the nuclear medicine physician - 18/12/2017

Following a similar structure of the previous interview, the nuclear medicine physician was interviewed. He showed that we can visualize the FDG PET scans based on what we are looking for. For instance, if we are looking for tumors, the scan is visualized based on the level of glucose.
The PET/CT fusion scan is a (2D) reconstruction (PET scan on top of the CT scan). The original data is the PET scan which show the 'shape' of the object. The anatomical references are needed to know exactly what is visualized.
Using Syngo by Siemens, the PET scans are visualized, usually in 4 windows (CT scan, PET/CT fusion scan, PET scan, PET scan (entire body)). Three main view modes: transverse plane (feet-head or head-feet), coronal plane (anterior-posterior or posterior-anterior) and lateral/sagittal plane (left-right or right-left). Often more than one screen can be used to show different kind of data at the same time (e.g., CT scan, PET scan and MRI scan). Other software used: Prism.

Writing the report is still an issue. Approximately, the 60% of an image assessment is spent for it. They contain clinical information, examination methods, results and CT conclusions (included specific measurements for follow-ups). At the moment, templates previously customized are used with automatic pre-text to standardize the reports and to avoid the daily-mood bias. This means that there is a limited (and most recurring) number of possible results/conclusions. However, free text is also available. In addition, default text is activated by key words (i.e., 'smart reporting'). Specific information and measurements have to be added manually.
In a similar way, we might prepare 'story templates' to facilitate the story creation process.
The radiologist found the concept of visual data stories interesting since it might motivate their findings (as a proof of what stated). By checking the provenance data patterns, one can assess whether some findings are due to coincidence, a systematic error, etc..

Furthermore, collaborative work performed without requiring human contact would let doctors/radiologists save much time and avoid bias due to human interaction.
Possible solution: to fill structured templates in.

NOTE: since the UMCG has already invested in the Epic (EMR system), our project might not be implemented there in the near future.